The relationship between algae and fungi that result in the formation of lichen is a classical example of mutualistic intermicrobial relationship (Figure 10.3).

Lichens are composed of primary producer, the phycosymbiont (algae) and a consumer the mycosymbiont (fungus)

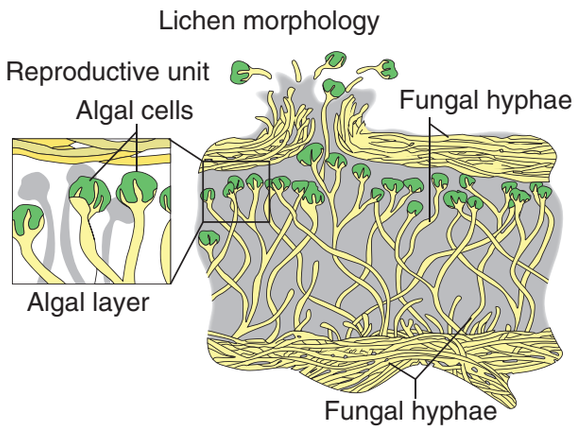
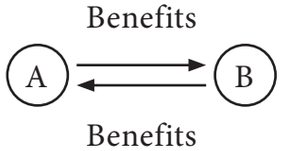


Figure 10.3: Lichen Morphology

Commensalism

In a commercial relationship between two microbial population, one population is benefited and other population remains unaffected. Commensalism is an unidirectional relationship between two population. The unaffected population does not benefit by the action of second population. For the receiving population, the benefit provided may be essential. In commensalism, the unaffected population modifies the habitat in such a way that another population is benefited.



For example: A population of facultative anaerobes utilizes oxygen and creates a habitat suitable for the growth of anaerobes. In soil, vitamin and growth factor producing organisms benefit vitamin and growth factors requiring organisms.

10.5.2 Harmful Microbial Interactions

Harmful microbial interaction is otherwise described as negative interaction or antagonistic interaction. Any inhibitory effect of an organism created by any means to the other organism is known as harmful interaction or antagonistic interaction and the phenomenon of this activity is called antagonism

Ammensalism

Ammensalism is the phenonmenon where one microbial species is affected by other species, where as other species is unaffected by first one. Ammensalism is accomplished by secretions of inhibitory substances such as antibiotics. Certain organisms may be of great practical importance, since they often produce antibiotics or other inhibitory substances, which affect the normal growth of other organisms. Antagonistic relationships are quite common in nature. For example: *Pseudomonas aeruginosa* is antagonistic towards *Aspergillus terreus* (Figure 10.4).

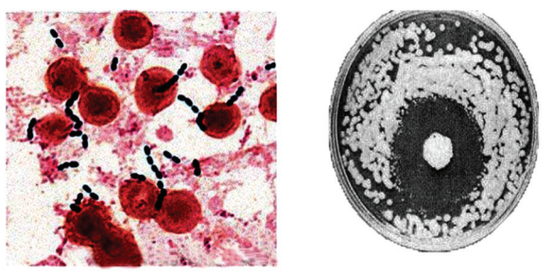


Figure 10.4: Microbial antagonism

Parasitism

This is a relationship in which one of the population benefits from the other and the host is usually harmed. Parasitism

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